📒 COMPLETE BIOLOGY NOTES

Ecology | Evolution | Biodiversity

🌿 PART 1: ECOLOGY

MODULE 1: INTRODUCTION TO ECOLOGY

Ecology is the scientific study of how organisms interact with each other and with their physical surroundings. The word comes from Greek — *oikos* (house) + *logos* (study). Ernst Haeckel coined it in 1866.

Levels of Ecological Organization

- **Organism** — A single living individual of any species
- **Population** — Group of individuals of the same species living in the same area at the same time
- **Community** — All the different species (populations) living together in one area
- **Ecosystem** — The community of living things plus all the non-living factors (water, soil, air, sunlight) in that area
- **Biome** — A large geographic region defined by its climate and dominant vegetation (e.g., desert, tundra)
- **Biosphere** — The entire zone of Earth where life exists — land, water, and atmosphere combined

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MODULE 2: ABIOTIC AND BIOTIC FACTORS

Abiotic Factors (Non-living)

These are the physical and chemical parts of the environment:

- **Temperature** — Controls metabolic rates; each species has a tolerance range
- **Water** — Most essential substance for life; availability determines where organisms live
- **Sunlight** — Primary energy source for photosynthesis; varies by latitude and season
- **Soil** — Provides nutrients and anchorage for plants; pH and mineral content matter

- **Wind** — Affects temperature, evaporation rates, and seed/pollen dispersal
- **Salinity** — Salt concentration in water; determines which aquatic organisms can survive

Biotic Factors (Living)

These are all the living components in an ecosystem:

- **Producers (Autotrophs)** — Make their own food through photosynthesis or chemosynthesis (plants, algae)
- **Consumers (Heterotrophs)** — Eat other organisms for energy (herbivores, carnivores, omnivores)
- **Decomposers** — Break down dead matter and recycle nutrients (bacteria, fungi)
- **Predators and Prey** — Hunting relationships that regulate population sizes
- **Parasites and Hosts** — One organism benefits at the expense of another
- **Competitors** — Organisms that compete for the same limited resources

Limiting Factors

A **limiting factor** is any resource that is in short supply and restricts population growth.

Liebig's Law of the Minimum states that growth is controlled not by the total amount of resources but by the scarcest resource. Even if everything else is abundant, one missing nutrient can stop growth entirely.

MODULE 3: POPULATION ECOLOGY

Population Characteristics

- **Population Size (N)** — Total number of individuals
- **Population Density** — Number of individuals per unit area or volume
- **Dispersion** — How individuals are spread across the area
- **Age Structure** — Proportion of individuals in different age groups (pre-reproductive, reproductive, post-reproductive)
- **Sex Ratio** — Proportion of males to females

Dispersion Patterns

- **Clumped** — Individuals cluster together; most common pattern; caused by resource availability or social behavior

- **Uniform** — Individuals are evenly spaced; caused by territorial behavior or competition
- **Random** — Individuals are scattered without pattern; rare in nature; occurs when resources are evenly distributed and there is no social interaction

Population Growth Models

Exponential Growth (J-curve)

- Formula: $dN/dt = rN$
- Occurs when resources are unlimited
- Population grows faster and faster without stopping
- Seen in bacteria in fresh medium, or organisms colonizing a new habitat
- The curve looks like the letter J

Logistic Growth (S-curve)

- Formula: $dN/dt = rN[(K-N)/K]$
- Occurs when resources are limited
- K = carrying capacity = maximum population the environment can sustain
- As population approaches K, growth slows down
- Four phases: Lag phase (slow start) → Exponential phase (rapid growth) → Deceleration phase (growth slows) → Equilibrium phase (population stabilizes near K)

Key Terms

- **r** = intrinsic rate of natural increase (birth rate minus death rate per individual)
- **K** = carrying capacity

Survivorship Curves

These describe the pattern of death across a species' lifespan:

- **Type I** — Low death rate early in life, most die old (humans, elephants)
- **Type II** — Constant death rate throughout life (birds, some reptiles)
- **Type III** — Very high death rate early in life, few survivors live long (oysters, fish, insects)

r-Selected vs K-Selected Species

r-selected species: - Produce many offspring - Little or no parental care - Short lifespan - Mature quickly - Thrive in unstable environments - Examples: insects, bacteria, rodents

K-selected species: - Produce few offspring - High parental care - Long lifespan - Mature slowly - Thrive in stable environments - Examples: humans, elephants, whales

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MODULE 4: COMMUNITY ECOLOGY

Species Interactions

- **Mutualism (+/+)** — Both species benefit (e.g., bee and flower)
- **Commensalism (+/0)** — One benefits, other is unaffected (e.g., barnacles on whale)
- **Parasitism (+/−)** — One benefits, other is harmed (e.g., tapeworm in human)
- **Competition (−/−)** — Both species are harmed by sharing limited resources
- **Predation (+/−)** — One organism kills and eats another

Competition

- **Intraspecific competition** — Between members of the same species (e.g., two deer competing for mates)
- **Interspecific competition** — Between members of different species (e.g., lions and hyenas competing for prey)

Competitive Exclusion Principle

Proposed by Gause. Two species competing for the exact same resources cannot coexist indefinitely. One will outcompete and eliminate the other. This is why species evolve to use different niches.

Resource Partitioning

When competing species divide up resources to reduce competition and coexist. For example, different warbler species feed at different heights in the same tree.

Ecological Niche

- **Fundamental niche** — The full range of conditions and resources a species could potentially use
- **Realized niche** — The actual range a species uses, limited by competition and other interactions
- Realized niche is always smaller than or equal to fundamental niche

Defense Mechanisms

- **Camouflage (Crypsis)** — Blending with surroundings to avoid detection (stick insect, leaf bug)

- **Aposematic coloring (Warning colors)** — Bright colors warn predators of toxicity (poison dart frog)
- **Batesian mimicry** — A harmless species mimics a harmful one (viceroy butterfly mimics monarch)
- **Müllerian mimicry** — Two or more harmful species resemble each other (bees and wasps both have yellow-black stripes)

Keystone Species

A species that has a disproportionately large effect on its ecosystem relative to its population size. Removing a keystone species causes major changes in the community. Example: sea otters control sea urchin populations, which protects kelp forests.

MODULE 5: ECOSYSTEMS

Energy Flow and Trophic Levels

Energy flows in one direction through an ecosystem — from sunlight to producers to consumers.

- **Trophic Level 1: Producers** — Capture solar energy through photosynthesis (plants, algae)
- **Trophic Level 2: Primary consumers** — Herbivores that eat producers (rabbits, grasshoppers)
- **Trophic Level 3: Secondary consumers** — Carnivores that eat herbivores (frogs, small birds)
- **Trophic Level 4: Tertiary consumers** — Top predators (hawks, lions)
- **Decomposers** — Work at all levels, breaking down dead organisms

10% Rule

Only about 10% of energy is transferred from one trophic level to the next. The remaining 90% is lost as heat through cellular respiration. This is why food chains rarely have more than 4-5 levels and why there are fewer top predators than herbivores.

Food Chain vs Food Web

- **Food chain** — A single linear pathway of energy transfer (grass → rabbit → fox → eagle)
- **Food web** — A complex network of interconnected food chains showing all feeding relationships in a community

Productivity

- **Gross Primary Productivity (GPP)** — Total amount of energy captured by producers through photosynthesis
- **Net Primary Productivity (NPP)** — Energy remaining after producers use some for their own respiration; NPP = GPP − Respiration
- NPP represents energy available to consumers

Productivity Across Ecosystems

- **High productivity** — Tropical rainforests, coral reefs, estuaries, wetlands
- **Low productivity** — Deserts, open ocean, tundra

MODULE 6: BIOGEOCHEMICAL CYCLES

These are the pathways through which chemical elements move between living organisms and the non-living environment.

Water Cycle (Hydrological Cycle)

- **Evaporation** — Water changes from liquid to vapor from surfaces
- **Transpiration** — Water vapor released from plant leaves
- **Condensation** — Water vapor cools and forms clouds
- **Precipitation** — Water falls as rain, snow, sleet, or hail
- **Runoff and infiltration** — Water flows over land or seeps into groundwater

Carbon Cycle

Carbon moves between atmosphere, organisms, oceans, and rocks.

Processes that add CO_2 to atmosphere: - Cellular respiration (all organisms) - Combustion of fossil fuels - Decomposition of dead matter - Volcanic eruptions

Processes that remove CO_2 from atmosphere: - Photosynthesis by plants and algae - Absorption by oceans - Formation of fossil fuels and sedimentary rocks (long-term storage)

Nitrogen Cycle

- Nitrogen gas (N_2) makes up 78% of the atmosphere but is unusable by most organisms
- **Nitrogen fixation** — Bacteria convert N_2 into ammonia (NH_3); done by Rhizobium in root nodules and some free-living bacteria

- **Nitrification** — Bacteria convert ammonia to nitrites and then nitrates (usable by plants)
- **Assimilation** — Plants absorb nitrates; animals get nitrogen by eating plants
- **Ammonification** — Decomposers break down dead organisms and waste, releasing ammonia
- **Denitrification** — Bacteria convert nitrates back to N_2, returning it to atmosphere

Phosphorus Cycle

- No atmospheric component — phosphorus does not enter the atmosphere
- Slowest biogeochemical cycle
- Stored in rocks and sediments
- Released by weathering of rocks
- Absorbed by plants from soil as phosphate
- Moves through food chains
- Returns to soil through decomposition
- Important for DNA, RNA, ATP, and bones

Eutrophication

- Excessive nutrients (nitrogen and phosphorus) enter water bodies from fertilizer runoff, sewage, or detergents
- Causes rapid algal growth (algal bloom)
- Algae block sunlight → aquatic plants die
- Dead algae decomposed by bacteria → oxygen depleted
- Results in dead zones where aquatic life cannot survive

MODULE 7: BIOMES

Terrestrial Biomes

Tropical Rainforest - Hot and wet year-round - Highest biodiversity of any biome - Dense canopy layers - Found near equator

Tropical Savanna - Warm with distinct wet and dry seasons - Grasslands with scattered trees - Large grazing animals and predators - Fire is a natural disturbance

Desert - Very low rainfall (less than 25 cm/year) - Extreme temperature variation between day and night - Organisms have water-conserving adaptations - Hot deserts and cold deserts both exist

Temperate Grassland - Moderate rainfall - Rich fertile soil - Few trees - Hot summers, cold winters - Examples: prairies, steppes

Temperate Deciduous Forest - Four distinct seasons - Trees lose leaves in autumn (deciduous) - Moderate rainfall throughout year

Taiga (Boreal Forest) - Cold climate with long winters - Dominated by coniferous trees (pine, spruce, fir) - Low biodiversity compared to tropical regions - Largest terrestrial biome

Tundra - Very cold, very little precipitation - Permafrost (permanently frozen soil layer) - No trees — only low shrubs, mosses, lichens - Short growing season - Found in Arctic regions and high mountain tops (alpine tundra)

Biome Distribution Factors

- **Latitude** — Distance from equator affects temperature
- **Altitude** — Higher elevation = colder temperatures (similar pattern to increasing latitude)
- **Temperature** — Major determinant of vegetation type
- **Precipitation** — Determines whether an area is forest, grassland, or desert

Aquatic Biomes

Freshwater Zones (Lakes and Ponds)

- **Littoral zone** — Shallow near-shore area; rooted plants; most biodiversity
- **Limnetic zone** — Open water where light penetrates; phytoplankton present
- **Profundal zone** — Deep water below light penetration; dark; low oxygen
- **Benthic zone** — Bottom of the water body; decomposers and bottom-dwelling organisms

Marine Zones (Oceans)

- **Intertidal zone** — Where ocean meets land; exposed at low tide, submerged at high tide; organisms must tolerate wave action and drying
- **Neritic zone** — Shallow ocean over continental shelf; high productivity; coral reefs found here
- **Oceanic zone** — Deep open ocean; low nutrient availability
- **Photic zone** — Upper layer where sunlight penetrates; photosynthesis occurs
- **Aphotic zone** — Deep water where no sunlight reaches; organisms rely on chemosynthesis or food falling from above

MODULE 8: ECOLOGICAL SUCCESSION

Succession is the gradual process of change in species composition of a community over time.

Primary Succession

- Occurs on surfaces where no soil exists (bare rock, new volcanic island, retreating glacier)
- Very slow process (can take hundreds of years)
- Pioneer species (lichens and mosses) colonize first
- Lichens break down rock → soil begins to form
- Gradually, grasses, shrubs, then trees establish
- Eventually reaches a stable climax community

Secondary Succession

- Occurs in areas where soil already exists but the community was disturbed (fire, flood, farming, logging)
- Faster than primary succession because soil and seed bank are already present
- Grasses and herbaceous plants colonize first
- Progresses through shrubs to trees

Key Terms

- **Pioneer species** — First organisms to colonize a disturbed or new area (lichens in primary, grasses in secondary)
- **Climax community** — Final stable community that persists until the next major disturbance
- **Seral stages** — The intermediate communities during the process of succession

MODULE 9: BIODIVERSITY AND CONSERVATION

Types of Biodiversity

- **Genetic diversity** — Variation in genes within a species; allows adaptation to changing environments
- **Species diversity** — Number of different species in an area
- **Ecosystem diversity** — Variety of different ecosystems in a region

Measuring Biodiversity

- **Species richness** — Simply the number of different species present
- **Species evenness** — How equally individuals are distributed among species; high evenness means no single species dominates

Threats to Biodiversity (HIPPO)

- **H — Habitat loss** — Deforestation, urbanization, agriculture; the biggest single threat
- **I — Invasive species** — Non-native species that outcompete or prey on native species
- **P — Pollution** — Chemicals, plastics, pesticides that harm organisms
- **P — Population growth** — More humans = more resource consumption and habitat destruction
- **O — Overexploitation** — Overhunting, overfishing, overharvesting beyond sustainable levels

Conservation Strategies

- **In-situ conservation** — Protecting species in their natural habitat (national parks, wildlife reserves, marine protected areas)
- **Ex-situ conservation** — Protecting species outside their natural habitat (zoos, aquariums, botanical gardens, seed banks, captive breeding programs)
- **Wildlife corridors** — Strips of habitat connecting isolated patches, allowing animals to move between them; reduces effects of habitat fragmentation

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MODULE 10: CLIMATE AND GLOBAL ECOLOGY

Greenhouse Effect

- **Natural greenhouse effect** — Greenhouse gases trap some heat in the atmosphere; essential for keeping Earth warm enough to support life
- **Enhanced greenhouse effect** — Human activities increase greenhouse gas concentrations → extra heat trapped → global warming

Greenhouse Gases

- **CO_2 (Carbon dioxide)** — From burning fossil fuels, deforestation; largest human contribution
- **CH_4 (Methane)** — From livestock, rice paddies, landfills; about 25 times more potent than CO_2 per molecule
- **N_2O (Nitrous oxide)** — From agriculture and fertilizers; strong greenhouse gas

- **CFCs (Chlorofluorocarbons)** — From refrigerants and aerosols; also destroy ozone layer

Effects of Climate Change

- **Species migration** — Organisms shift to higher latitudes and altitudes
- **Seasonal shifts** — Earlier springs, later autumns; mismatches in food availability and breeding timing
- **Coral bleaching** — Warm water causes corals to expel symbiotic algae; leads to death
- **Ocean acidification** — CO_2 dissolves in ocean → forms carbonic acid → lowers pH → harms shell-forming organisms
- **Sea level rise** — Melting ice caps and thermal expansion of water → flooding of coastal habitats

Key Formulas for Ecology

- Growth rate: $r = (\text{births} - \text{deaths}) / N$
- Exponential growth: $dN/dt = rN$
- Logistic growth: $dN/dt = rN[(K-N)/K]$
- Population density: $D = N / \text{Area}$
- Net Primary Productivity: NPP = GPP − Respiration

🧬 PART 2: EVOLUTION

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1. WHAT IS EVOLUTION?

Evolution is the change in allele frequencies in a population over successive generations. It is populations that evolve, not individual organisms. An individual may be selected for or against, but the genetic makeup of the population changes over time.

Key Terms

- **Population** — A group of individuals of the same species living in the same area and interbreeding
- **Gene pool** — The total collection of all alleles (gene variants) present in a population

- **Allele frequency** — The proportion of a particular allele among all copies of that gene in the population

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2. HISTORY OF EVOLUTIONARY THOUGHT

Charles Darwin

- Traveled on HMS Beagle; observed variation in Galápagos finches
- Proposed the theory of evolution by **natural selection**
- Published *On the Origin of Species* (1859)

Darwin's Four Postulates

1. **Variation exists** — Individuals in a population differ in their traits
2. **Heritability** — Some of these variations are passed from parents to offspring
3. **Overproduction** — More offspring are produced than can survive
4. **Differential survival and reproduction** — Individuals with favorable traits are more likely to survive and reproduce

The result is **descent with modification** — populations change over time as beneficial traits become more common.

Alfred Russel Wallace

- Independently developed a similar theory of natural selection
- His work prompted Darwin to publish

Lamarckism

- Jean-Baptiste Lamarck proposed that organisms could pass on traits acquired during their lifetime
- Example: a blacksmith's children would be born with strong arms
- This is incorrect — acquired traits are not inherited
- Important historically because it was an early attempt to explain evolution

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3. MECHANISMS OF EVOLUTION

There are four main mechanisms that cause evolution:

3.1 Natural Selection

The process by which individuals with traits better suited to their environment survive and reproduce more successfully.

Types of Natural Selection:

1. **Directional Selection** - Favors one extreme phenotype - Shifts the population's trait distribution in one direction - Example: Longer necks in giraffes favored for reaching higher leaves

2. **Stabilizing Selection** - Favors the average/intermediate phenotype - Reduces variation in the population - Example: Human birth weight — very large or very small babies have lower survival

3. **Disruptive Selection** - Favors both extreme phenotypes over the average - Increases variation; can split a population - Can lead to speciation - Example: Birds with very large or very small beaks survive better than medium-beak birds

4. **Fitness** - Biological fitness = an organism's ability to survive AND reproduce - It is not about physical strength - Measured by the number of fertile offspring an organism produces

5. **Adaptation** - A heritable trait that increases an organism's fitness in its environment - Result of natural selection over many generations - Examples: thick fur in arctic animals, long roots in desert plants

6. **Sexual Selection** - A form of natural selection where traits are favored because they increase mating success - **Intersexual selection** — One sex chooses mates based on traits (female peacocks choose males with largest tails) - **Intrasexual selection** — Members of one sex compete for mates (male deer fighting with antlers)

3.2 Mutation

- A random, permanent change in the DNA sequence
- Types include point mutations, insertions, deletions, duplications
- Most mutations are neutral; some are harmful; rarely, some are beneficial
- **Mutation is the only source of entirely new genetic variation**
- Without mutation, there would be no raw material for evolution
- Mutations are random with respect to fitness — they do not occur because an organism needs them

3.3 Genetic Drift

- Random changes in allele frequencies due to chance events
- Has the strongest effect in **small populations**
- Can cause alleles to become fixed (100%) or lost (0%) by chance alone
- Does not produce adaptation — changes are random

Bottleneck Effect - A catastrophic event (disease, natural disaster) drastically reduces population size - Surviving population has reduced genetic diversity - Allele frequencies in survivors may differ greatly from original population - Example: Northern elephant seals were hunted to near extinction; surviving population has very low genetic diversity

Founder Effect - A small group of individuals colonizes a new area and establishes a new population - The new population's gene pool is only a sample of the original population - Can lead to very different allele frequencies - Example: Amish communities have high frequency of certain genetic disorders due to small founding group

3.4 Gene Flow

- The movement of alleles from one population to another through migration of individuals
- Increases genetic diversity within a population
- Makes populations more similar to each other
- Can introduce new alleles or change existing allele frequencies
- Counteracts the effects of genetic drift and natural selection that make populations diverge

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4. HARDY-WEINBERG EQUILIBRIUM

A mathematical model describing a population that is NOT evolving.

The Equations

- Allele frequencies: $\mathbf{p + q = 1}$
  - p = frequency of dominant allele
  - q = frequency of recessive allele
- Genotype frequencies: $\mathbf{p^2 + 2pq + q^2 = 1}$
  - p^2 = frequency of homozygous dominant (AA)
  - 2pq = frequency of heterozygous (Aa)
  - q^2 = frequency of homozygous recessive (aa)

Five Conditions Required

For a population to be in Hardy-Weinberg equilibrium (not evolving), ALL of these must be true:

1. **No mutation —** No new alleles are created
2. **No migration (gene flow)** — No individuals enter or leave
3. **No natural selection** — All genotypes have equal fitness
4. **Random mating —** No preference for certain genotypes
5. **Large population size** — No genetic drift

If any condition is violated, the population is evolving.

How to Use It

- Usually, you are given the percentage of homozygous recessive individuals
- That percentage = q^2
- Take the square root to find q
- Then $p = 1 - q$
- Then calculate p^2, 2pq as needed

Practical Use

- Comparing observed genotype frequencies to expected H-W frequencies can reveal whether evolution is occurring at a particular gene

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5. SPECIATION AND MACROEVOLUTION

Biological Species Concept

A species is a group of organisms that can interbreed in nature and produce fertile offspring. This concept has limitations (cannot apply to asexual organisms or fossils).

Speciation

The process by which one species splits into two or more separate species.

Allopatric Speciation - Caused by geographic isolation (physical barrier separates populations) - Barrier examples: mountain range, river, ocean, glacier - Separated populations experience different mutations, selection pressures, and drift - Over time, they become so different they can no longer interbreed - Example: Darwin's finches on different Galápagos islands

Sympatric Speciation - Occurs without geographic separation — within the same area - Mechanisms include: - **Polyploidy —** Organism has extra sets of chromosomes; common in plants; polyploid individuals cannot breed with diploid ancestors - **Habitat differentiation —** Populations adapt to use different resources in the same area - **Behavioral isolation —** Different mating calls or rituals evolve - Less common than allopatric speciation

Reproductive Isolation

Mechanisms that prevent gene flow between species:

Prezygotic barriers (before fertilization) - **Temporal isolation —** Species breed at different times (different seasons or times of day) - **Habitat isolation —** Species live in different habitats in the same area - **Behavioral isolation —** Different courtship rituals or signals - **Mechanical isolation —** Physical incompatibility of reproductive structures - **Gametic isolation —** Sperm and egg cannot fuse

Postzygotic barriers (after fertilization) - **Hybrid inviability —** Hybrid embryo does not develop properly - **Hybrid sterility —** Hybrid is viable but cannot reproduce (e.g., mule = horse × donkey) - **Hybrid breakdown —** First generation hybrid is fertile but subsequent generations have reduced fitness

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6. EVIDENCE FOR EVOLUTION

Fossil Record

- Fossils show how organisms have changed over time
- Transitional fossils show intermediate forms between ancestral and modern species
- Examples: Archaeopteryx (between dinosaurs and birds), Tiktaalik (between fish and amphibians)
- Older fossils are found in deeper rock layers

Comparative Anatomy

Homologous Structures - Same underlying structure, different function - Inherited from a common ancestor - Evidence of divergent evolution - Example: Human arm, whale flipper, bat wing, cat leg — all have the same bone arrangement

Analogous Structures - Similar function but different underlying structure - Result of convergent evolution (similar environmental pressures) - NOT evidence of common ancestry - Example: Bird wings and insect wings both enable flight but are structurally different

Vestigial Structures - Reduced or functionless structures that were functional in ancestors - Examples: Human appendix, whale pelvic bones, snake leg bones

Embryology

- Embryos of related species look very similar in early development
- Vertebrate embryos all show gill slits, tails, and similar structures early on
- Suggests common ancestry

Biogeography

- Geographic distribution of species reflects evolutionary history
- Island species are similar to nearby mainland species (descended from mainland colonizers)
- Similar environments on different continents have different species that independently evolved similar adaptations (convergent evolution)

Molecular Biology

- DNA and protein sequences can be compared between species
- More similar sequences indicate more recent common ancestry
- All life shares the same genetic code — evidence of universal common ancestry
- Molecular clocks use mutation rates to estimate when species diverged

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7. PHYLOGENY

Phylogenetic Trees

Diagrams that show evolutionary relationships among species.

- **Node** — A branching point; represents the most recent common ancestor of the groups that diverge from it
- **Branch** — A line representing a lineage through time
- **Clade** — A group consisting of an ancestor and all its descendants (a monophyletic group)
- **Root** — The base of the tree; represents the most ancient common ancestor
- **Sister taxa** — Two groups that share an immediate common ancestor

How to Read a Tree

- The pattern of branching shows relatedness, not the position on the tips
- Species that share a more recent node are more closely related
- Trees can be rotated at any node without changing the relationships

Classification

- Traditional classification: Domain → Kingdom → Phylum → Class → Order → Family → Genus → Species
- Modern classification increasingly uses phylogenetic (cladistic) approaches based on evolutionary relationships

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8. EVOLUTION IN ACTION

Antibiotic Resistance

- When antibiotics are used, most bacteria die but resistant individuals survive
- Resistant bacteria reproduce and pass resistance genes to offspring
- Overuse and misuse of antibiotics accelerate this process
- A major public health concern

Pesticide Resistance

- Similar to antibiotic resistance
- Insects with resistance genes survive pesticide application
- Their offspring inherit resistance
- Over generations, the pest population becomes resistant

Viral Evolution

- Viruses mutate rapidly, especially RNA viruses
- This is why new flu vaccines are needed each year
- HIV mutates so quickly that it is difficult to develop a permanent cure
- COVID-19 variants arose through viral evolution

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9. APPLICATIONS OF EVOLUTION

Medicine

- Understanding how pathogens evolve helps develop better drugs and vaccines
- Pharmacogenomics uses evolutionary principles to tailor treatments
- Cancer is an evolutionary process within the body (cells mutate and are selected)

Agriculture

- Artificial selection has produced all domesticated crops and livestock
- Understanding pest evolution helps design sustainable pest management

- Genetic diversity in crops provides insurance against disease

Conservation Biology

- Maintaining genetic diversity is essential for species survival
- Small populations lose diversity through genetic drift
- Conservation genetics guides breeding programs and habitat management

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🌍 PART 3: BIODIVERSITY

MODULE 1: WHAT IS LIFE AND ORIGIN OF LIFE

Characteristics of Living Things

All living organisms share these features:

- **Cellular organization** — All life is made of one or more cells
- **Metabolism** — Chemical reactions that convert energy and build molecules
- **Growth and development** — Organisms increase in size and complexity following genetic instructions
- **Reproduction** — Organisms produce offspring (sexual or asexual)
- **Response to stimuli** — Organisms react to changes in their environment
- **Evolutionary adaptation** — Populations change over generations in response to their environment

Origin of Life Theories

Abiogenesis - Life arose from non-living matter through natural chemical processes - Supported by Miller-Urey experiment (1953) — simulated early Earth atmosphere and produced amino acids

Primordial Soup Theory - Early Earth’s oceans contained simple organic molecules - Energy from lightning, UV radiation, and volcanic activity drove chemical reactions - Simple molecules combined to form more complex ones (amino acids, nucleotides)

Hydrothermal Vent Hypothesis - Life may have originated near deep-sea hydrothermal vents - Vents provide heat, minerals, and chemical energy - Some of Earth’s oldest fossils are found near ancient vent sites

Common Ancestry

- All living organisms share a common ancestor (Last Universal Common Ancestor — LUCA)
- Evidence: universal genetic code, shared biochemistry, similar cellular structures
- Biodiversity is the result of billions of years of evolution from this common origin

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MODULE 2: PROKARYOTIC LIFE

What Are Prokaryotes?

- Single-celled organisms without a membrane-bound nucleus
- DNA is in a nucleoid region (not enclosed in a nuclear membrane)
- Generally smaller and simpler than eukaryotic cells
- Reproduce by binary fission (asexual)
- Can reproduce very rapidly

Two Domains of Prokaryotes

Bacteria - Most familiar prokaryotes - Found everywhere — soil, water, air, inside other organisms - Cell wall contains peptidoglycan - Shapes: cocci (spherical), bacilli (rod), spirilla (spiral)

Archaea - Superficially similar to bacteria but biochemically distinct - Cell wall lacks peptidoglycan - Often found in extreme environments (extremophiles): hot springs, salt lakes, deep ocean vents - Also found in normal environments like soil and oceans - More closely related to eukaryotes than to bacteria

Ecological Importance of Prokaryotes

- **Nitrogen fixation** — Certain bacteria convert atmospheric N_2 to usable forms
- **Decomposition** — Break down dead matter and recycle nutrients
- **Human microbiome** — Trillions of bacteria in our gut aid digestion, produce vitamins, and protect against pathogens
- **Biogeochemical cycling —** Essential in carbon, nitrogen, sulfur, and phosphorus cycles

Pathogenic Bacteria and Antibiotics

- Some bacteria cause disease (tuberculosis, cholera, strep throat)
- Antibiotics are drugs that kill or inhibit bacteria
- Overuse leads to antibiotic-resistant bacteria (MRSA, XDR-TB)

- Resistance is an example of natural selection in action

MODULE 3: EUKARYOTIC LIFE

What Are Eukaryotes?

- Cells with a membrane-bound nucleus containing DNA
- Have membrane-bound organelles (mitochondria, endoplasmic reticulum, Golgi apparatus)
- Generally larger and more complex than prokaryotes
- Include both unicellular and multicellular organisms

Endosymbiotic Theory

- Mitochondria and chloroplasts were once free-living prokaryotes
- They were engulfed by ancestral eukaryotic cells and became organelles
- Evidence: both have their own DNA, double membranes, and divide independently

Major Groups of Eukaryotes

Protists - Mostly unicellular eukaryotes (some multicellular like seaweed) - Very diverse group — not a true evolutionary group (paraphyletic) - Includes: - Plant-like protists (algae) — photosynthetic - Animal-like protists (protozoa) — heterotrophic (amoeba, paramecium) - Fungus-like protists (slime molds) - Important as base of aquatic food chains (phytoplankton)

Fungi - Mostly multicellular (yeasts are unicellular) - Cell walls made of chitin - Heterotrophic — absorb nutrients from surroundings - Major decomposers — break down dead organic matter - Reproduce by spores - Examples: mushrooms, molds, yeasts - Some form mutualistic relationships: - Mycorrhizae — fungi + plant roots (help plants absorb water and minerals) - Lichens — fungi + algae/cyanobacteria

Plants - Multicellular, photosynthetic eukaryotes - Cell walls made of cellulose - Produce oxygen and form the base of most terrestrial food chains - Major groups: mosses, ferns, gymnosperms (conifers), angiosperms (flowering plants)

Prokaryotes vs Eukaryotes Summary

Feature	Prokaryotes	Eukaryotes
Nucleus	Absent	Present
Organelles	Few/none membrane-bound	Many membrane-bound
Size	Smaller (1-10 μm)	Larger (10-100 μm)
DNA	Circular	Linear chromosomes
Reproduction	Binary fission	Mitosis/meiosis
Complexity	Simple	Complex

MODULE 4: INVERTEBRATE ANIMALS

What Are Animals?

- Multicellular, heterotrophic eukaryotes
- No cell walls
- Most can move
- Evolved from colonial protist ancestors
- Over 95% of animal species are invertebrates (no backbone)

Body Plan Concepts

- **Symmetry**: Radial (circular — jellyfish) vs Bilateral (left-right mirror — insects, humans)
- **Body cavity**: Acoelomate (no cavity) vs Coelomate (true body cavity)
- **Germ layers**: Diploblastic (two layers) vs Triploblastic (three layers)

Major Invertebrate Groups

Porifera (Sponges) - Simplest animals - No true tissues - Filter feeders - Sessile (attached to substrate)

Cnidarians - Radial symmetry - Two body forms: polyp (attached) and medusa (free-swimming) - Stinging cells (cnidocytes) for capturing prey - Examples: jellyfish, corals, sea anemones

Platyhelminthes (Flatworms) - Bilateral symmetry - Flat body - Some are parasitic (tapeworms, flukes)

Nematoda (Roundworms) - Cylindrical body - Found in almost every habitat - Many are parasitic

Annelida (Segmented Worms) - Segmented body - Examples: earthworms, leeches

Mollusks - Soft body, often with shell - Muscular foot for movement - Examples: snails, clams, octopus, squid

Arthropods - Largest animal phylum - Exoskeleton made of chitin - Jointed appendages - Segmented body - Includes: insects, spiders, crustaceans, centipedes - Insects are the most diverse group of organisms on Earth

Echinoderms - Marine only - Radial symmetry (as adults) - Water vascular system - Examples: starfish, sea urchins

Ecological Importance of Invertebrates

- Pollination (insects)
- Decomposition (worms, insects)
- Food source for other animals
- Coral reefs (cnidarians) support enormous biodiversity
- Soil health (earthworms)

MODULE 5: VERTEBRATE ANIMALS

Key Feature

All vertebrates have a vertebral column (backbone) that protects the spinal cord. They belong to phylum Chordata, subphylum Vertebrata.

Major Vertebrate Groups

Fish - First vertebrates to evolve - Aquatic; breathe through gills - Ectothermic (cold-blooded) - Three groups: - Jawless fish (lampreys, hagfish) - Cartilaginous fish (sharks, rays) - Bony fish (most fish species — salmon, goldfish)

Amphibians - Live part of life in water, part on land - Ectothermic - Moist skin used for gas exchange - Eggs laid in water (no shell) - Undergo metamorphosis (tadpole → adult) - Examples: frogs, toads, salamanders

Reptiles - Fully terrestrial (most species) - Ectothermic - Dry scaly skin prevents water loss - Amniotic eggs with shell (can be laid on land) - Examples: snakes, lizards, turtles, crocodiles

Birds - Endothermic (warm-blooded) - Feathers for insulation and flight - Lightweight hollow bones - High metabolic rate - Lay hard-shelled eggs - Evolved from theropod dinosaurs - Examples: eagles, penguins, sparrows

Mammals - Endothermic - Hair or fur for insulation - Mammary glands produce milk to feed young - Most have live birth - Highly developed brain - Three groups: - Monotremes — Egg-laying mammals (platypus, echidna) - Marsupials — Pouched mammals; young born undeveloped (kangaroo, koala) - Placental mammals — Young develop fully inside uterus with placenta (humans, whales, bats)

MODULE 6: MAMMALS AND HUMANS

Primate Characteristics

- Forward-facing eyes (binocular vision and depth perception)
- Opposable thumbs (grasping ability)
- Large brain relative to body size
- Complex social behavior
- Extended parental care

Primate Groups

- Prosimians (lemurs, lorises)
- New World monkeys (spider monkeys — prehensile tails)
- Old World monkeys (baboons — no prehensile tails)
- Apes (gorillas, chimpanzees, orangutans, gibbons, humans)

Human Evolution

- Humans belong to genus *Homo*, species *Homo sapiens*
- Evolved from primate ancestors in Africa
- Key milestones:
  - **Bipedalism** — Walking upright on two legs (earliest hominins, ~6-7 million years ago)
  - **Tool use** — Stone tools (~2.5 million years ago)
  - **Brain enlargement** — Progressive increase in cranium size
  - **Language and culture** — Complex communication and social structures

- Humans did not evolve from modern apes — humans and apes share a common ancestor
- Closest living relatives: chimpanzees and bonobos

BIODIVERSITY CORE CONCEPTS

What Is Biodiversity?

The variety of life on Earth at all levels of organization.

Three Levels

- **Genetic diversity** — Variety of genes and alleles within a species; higher diversity = better ability to adapt
- **Species diversity** — Number and variety of species in an area
- **Ecosystem diversity** — Range of different habitats, communities, and ecological processes in a region

Patterns of Biodiversity

- Biodiversity is highest near the equator and decreases toward the poles (latitudinal gradient)
- Larger areas support more species (species-area relationship)
- Islands have fewer species than mainlands of equal area but often have unique endemic species
- Tropical rainforests and coral reefs have the highest species diversity

Importance of Biodiversity

- **Ecosystem services** — Clean air, water purification, pollination, climate regulation
- **Food and agriculture** — Wild relatives of crops provide genetic resources
- **Medicine** — Many drugs derived from natural organisms
- **Economic value** — Tourism, fisheries, timber, agriculture
- **Ethical and aesthetic value** — Intrinsic right of species to exist; beauty and cultural importance

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THREATS TO BIODIVERSITY

HIPPO Framework

- **Habitat destruction** — Deforestation, urbanization, agriculture converting natural habitats; greatest single threat to biodiversity
- **Invasive species** — Non-native species introduced to new areas that outcompete, prey on, or bring disease to native species
- **Pollution** — Pesticides, industrial waste, plastics, oil spills; affects air, water, and soil quality
- **Population growth (human)** — More people means more demand for resources, more habitat conversion, more waste
- **Overexploitation** — Taking species faster than they can replenish (overfishing, poaching, illegal wildlife trade)

Additional Threats

- **Climate change** — Alters habitats, shifts ranges, disrupts timing of ecological events
- **Disease** — Emerging diseases can devastate populations (chytrid fungus in amphibians)

CONSERVATION OF BIODIVERSITY

In-situ Conservation

- Protecting species in their natural habitats
- National parks, wildlife sanctuaries, biosphere reserves, marine protected areas
- Most effective form of conservation because it preserves the entire ecosystem

Ex-situ Conservation

- Protecting species outside their natural habitats
- Zoos, aquariums, botanical gardens
- Seed banks, gene banks
- Captive breeding programs with potential for reintroduction

Other Strategies

- **Wildlife corridors** — Connect fragmented habitats so animals can move between them
- **Legislation** — Laws like the Endangered Species Act, CITES (Convention on International Trade in Endangered Species)
- **Community involvement** — Local communities participate in conservation efforts

- **Sustainable use** — Using resources at a rate that allows natural replenishment

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QUICK REFERENCE TABLE

Topic	**Key Takeaway**
Life characteristics	Cells, metabolism, reproduction, response, growth, evolution
Prokaryotes	No nucleus; Bacteria and Archaea
Eukaryotes	Nucleus present; Protists, Fungi, Plants, Animals
Invertebrates	No backbone; 95%+ of animal species
Vertebrates	Backbone; Fish, Amphibians, Reptiles, Birds, Mammals
Mammals	Hair, milk, warm-blooded
Human evolution	Bipedalism, tool use, brain growth, culture
Biodiversity types	Genetic, Species, Ecosystem
Threats	HIPPO — Habitat loss, Invasive species, Pollution, Population, Overexploitation
Conservation	In-situ (natural habitat) and Ex-situ (zoos, banks)

FINAL CORE PRINCIPLES

- All life shares a common ancestor and is connected through evolution
- Biodiversity is the result of billions of years of evolutionary processes
- Ecology describes how living things interact with each other and their environment
- Evolution is the mechanism that generates and shapes biodiversity
- Human activities are the primary current threat to biodiversity
- Conservation requires understanding ecology, evolution, and biodiversity together

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